

1. Class and Member Function (Inside Class)

Question: Define a class `student` with public data members `name` (string) and `rollNo` (int). Write a member function `display()` inside the class that prints the student's details. In `main()`, create one object and call `display()`.

2. Member Function Defined Outside the Class

Question: Create a class `circle` with a private data member `radius` (double). Declare a member function `area()` inside the class, but define it outside the class. In `main()`, create an object, set radius, and print the area using `area()`.

3. Objects as Function Arguments

Question: Define a class `Time` with `hours` and `minutes` (int). Write a function `addTime(Time t1, Time t2)` that takes two `Time` objects as arguments, adds their hours and minutes separately, and prints the total time. Call this function from `main()`.

4. Constructors – Parameterized Constructor

Question: Write a class `Box` with `length`, `breadth`, `height` (int). Create a parameterized constructor to initialize these values. In `main()`, create a `Box` object by passing values (3,4,5) and calculate its volume.

5. Overloaded Constructors

Question: Define a class `Fraction` with numerator and denominator (int). Provide three constructors:

- Default (sets to 1/1)
 - One parameter (sets numerator to given value, denominator to 1)
 - Two parameters (sets both)
- In `main()`, create objects using each constructor and print the fraction values.

6. Default Copy Constructor

Question: Create a class `Point` with `x` and `y` (int). Write a parameterized constructor. In `main()`, create one `Point` object `p1` (2,3). Then create another object `p2` by copying `p1` using the default copy constructor (`Point p2 = p1;`). Print both points to show they are identical.

7. Destructor

Question: Write a class `Counter` that increments a static variable `count` in the constructor and decrements it in the destructor. Also provide a static function `getCount()`. In `main()`, create three objects inside a block `{ }`, print count after each creation, and then see the count decrease when objects go out of scope.

8. Static Class Data Member

Question: Define a class `BankAccount` with a static data member `interestRate` (double) and a regular member `balance` (double). Write a static function `setInterestRate(double)` to change the rate and a non-static function `calculateInterest()` that returns `balance * interestRate`. In `main()`, set rate to 7.5%, create two accounts with different balances, and print their interests.

9. Single Inheritance (Base and Derived Class)

Question: Create a base class `Vehicle` with a public member function `start()` that prints "Vehicle started". Derive a class `Car` from `Vehicle`. In `Car`, add a member function `honk()` that prints "Beep beep!". In `main()`, create a `Car` object and call both `start()` and `honk()`.

10. Objects as Arguments (Revisited – by Reference)

Question: Define a class `Distance` with `feet` and `inches` (int). Write a function `addDist(Distance &d1, Distance &d2)` that takes two `Distance` objects by reference, adds them (adjusting inches to feet if ≥ 12), and returns a new `Distance` object. Call this function from `main()` and print the result.